Pegasus Researcher Guide to Resource Allocation

Introduction

This guide explains the changes researchers can expect from the SLURM resource manager upgrade on Pegasus, scheduled to start on February 7th, 2026. During the initial transition period, most Pegasus partitions are intentionally configured to preserve legacy whole-node behavior while researchers adjust to new submission requirements. Noteworthy changes apply to:

1. `sbatch` scripts,
2. `salloc`, and
3. `srun`.

All other SLURM commands will function as before. We are aware some of these changes may seem different at first, but Research Technology Services (RTS) is ready to assist you during this important upgrade. Please email any questions you have during this process to rtshelp@gwu.edu.

What changed?

Pegasus now uses trackable resources (TRES) as its scheduling policy.[1] TRES treats CPUs, memory, and GPUs as independent resources which shortens times and increases job throughput. These benefits require a small number of *best practice* changes to the way jobs are submitted. Previously, allocations reserved entire nodes, resulting in suboptimal utilization.

Four Things Researchers Should Know

1. Simplified Partitions
2. Memory Allocation is Now Enforced
3. GPU Jobs Must Request GPUs Explicitly
4. Explicit Resource Requests are Now Enforced

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[1] *Trackable resources became SLURM's default scheduling policy as of 23.11.*

1. Simplified Partitions

Previously, Pegasus had nearly 30 partitions that divided the cluster based on GPU, memory, core count, or a dedicated purpose (high throughput, debug, nano, etc). The result was:

- A complex and segmented partition layout
- Researchers overloading "`--partition`" as a scheduling hint rather than a policy control
- Long queues while other partitions remained underutilized
- Misallocation of high-value hardware

With resource-centric allocation, most legacy partitions are no longer needed. The new streamlined partition layout will be as follows:

New Partition Definitions		
	Partition	**Purpose**
1	cpu	Non-GPU workflows
2	gpu	All GPU workflows
3	longJob	Workflows requiring up to 42 days
4	viz	Dedicated visualization workflows
5	aws*	Cloud Burst/Burst Buffers
6	basestar	Grace Hoppers

Researchers submitting CPU-based jobs can often ignore the `--partition` option altogether. A researcher submitting a GPU-based job **must** include `--partition=gpu`. See **Section 3** for more information.

NOTE
To facilitate the transition to TRES scheduling, legacy Pegasus partitions will remain in place for 90 days after the cutover, until roughly the end of the Spring 2026 semester. We strongly encourage researchers to adopt the new partition scheme as soon as possible, in order to avoid issues when the transition period ends.

Additionally, **during this transition period, CPU-based jobs will need to specify the "cpu" partition.** This won't be the case after the old partitions are fully retired.

2. Memory Allocation is Now Enforced

Memory usage is now strictly enforced using cgroups/v2. Jobs that explicitly request memory will be limited to that requested amount. Jobs that do not request memory will receive a default allocation based on the number of allocated CPU cores.

NOTE
If more memory is requested than the 16 GB per core limit allowed, then the scheduler will automatically assign additional cores. Note that for hcc nodes (384 cores), the maximum physical memory (2.3 TB) available is 6 GB per core.

Recommended:

```
#SBATCH --mem=32G
```

OR

```
#SBATCH --mem-per-cpu=4G
```

Jobs that previously exceeded requested memory without failing may now terminate earlier and more predictably.

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3. GPU Jobs Must Request GPUs Explicitly

On legacy Pegasus partitions, GPU access may still be implicitly granted to preserve existing workflows during the transition period. However, on the new simplified partitions, GPUs must be explicitly requested using `--gres`. **Researchers are strongly encouraged to explicitly request GPUs from hereon in all jobs that need GPU access, in order to ensure portability and future compatibility.** This is especially important when considering that the legacy partitions will be phased out, thus all GPU jobs will require explicitly requesting GPUs at that point.

Required for all GPU jobs:

```
#SBATCH --partition=gpu
#SBATCH --gres=gpu:<TYPE>:1
```

Jobs without “`--gres`” defined will run without GPU access and will most likely fail as a result.

> **TIP**
> The <TYPE> of GPU (and number per node) can be found from either of the commands below:
>
> ```
> scontrol show nodes | grep -i gres
> sinfo -o "%P %G"
> ```

Optional (but recommended) SLURM option configuration for GPU workflows:

```
#SBATCH --cpus-per-gpu=4
#SBATCH --mem-per-gpu=16G
```

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4. Explicit Resource Requests are Now Enforced

SLURM now enforces CPU, memory, and GPU requests exactly as specified in job scripts. Jobs that do not explicitly request resources will continue to receive default allocations during the transition period.

Although SLURM supports safe node sharing, most Pegasus partitions are currently configured to preserve legacy whole-node behavior. Node sharing will be enabled more broadly once the transition period is finished.

If a workflow requires exclusive access to a node, researchers may still request it explicitly:

```
#SBATCH --exclusive
```

That being said, researchers are encouraged to request only the resources their jobs require, as this improves scheduling efficiency and overall throughput.

Common Examples

CPU-only Job

Allocate 4 CPU cores and 8 GB of memory for 1 hour. *(Note: Once the transition period has finished around mid-May 2026, the partition option no longer needs to be explicitly defined.)*

```
#SBATCH --time=0-01:00:00
#SBATCH --partition=cpu
#SBATCH --cpus-per-task=4
#SBATCH --mem=8G
```

Single GPU Job

Allocate 1 GPU of type <TYPE>, 4 CPU cores, and 32 GB of memory for 1 hour.

```
#SBATCH --time=0-01:00:00
#SBATCH --partition=gpu
#SBATCH --gres=gpu:<TYPE>:1
#SBATCH --cpus-per-gpu=4
#SBATCH --mem-per-gpu=32G
```

Multi-GPU Job

Allocate 4 GPUs of type <TYPE>, 24 CPU cores, and 96 GB of memory for 1 hour.

```
#SBATCH --time=0-01:00:00
#SBATCH --partition=gpu
#SBATCH --gres=gpu:<TYPE>:4
#SBATCH --cpus-per-gpu=6
#SBATCH --mem-per-gpu=24G
```

What should I do if my job fails?

Any job failures after the upgrade are more likely to be caused by missing or insufficient CPU, memory, or GPU requests, rather than scheduler errors. If your job fails, runs slowly, doesn't properly start, or otherwise behaves unexpectedly, we suggest first checking the following:

1. Check the output (.out) and error (.err) files for warnings and error messages.
2. If this was a GPU job, were GPUs explicitly requested?
3. Was the number of requested CPU cores / tasks sufficient?
4. Was too much or too little memory requested? Recall, the number of allocated cores scales with requested memory.

TIP
The answers to questions 2-4 may all be found with the following command, where `<jobid>` is your job ID within SLURM:

```
sacct -X -j <jobid> -o ReqTRES%50,AllocTRES%50
```

If you would like to discuss how this change may affect your workflows, please contact rtshelp@gwu.edu or join one of our Office Hours every Tuesday and Thursday from 12:30–2:30 PM. Zoom Link

If a job fails after the upgrade, please include the job ID and submission script when contacting support. We are happy to help update scripts or explain new behavior.